Sky Motion, Seasons, the Moon and Eclipses

Covers Lectures 1.1–1.3 about 75 minutes

Name: _____ Date: _____ Marks: _____ / 70

You may use: a basic calculator, your lecture notes and the declination table from Lecture 1.2. **Difficulty:** ★ remember ★★ apply ★★★ challenge (bonus)

Section A — Fill in the blanks ★ 10 marks

1. Earth rotates from _____ to _____, so the sky appears to turn from east to west.
2. The sky appears to turn by _____° every hour, or 1° every _____ minutes.
3. The altitude of Polaris (Dhruva) above the northern horizon is roughly equal to your _____.
4. A sidereal day lasts _____ h _____ min, about _____ minutes shorter than a solar day.
5. Earth's axis is tilted by about _____° to the perpendicular of its orbit.
6. Earth is closest to the Sun around _____ (month); this point is called _____.
7. The apparent yearly path of the Sun among the stars is called the _____.
8. The Tropic of Cancer passes through _____ Indian states.
9. A solar eclipse can happen only on _____ (tithi), and a lunar eclipse only on _____.
10. In Indian astronomy the ascending node of the Moon's orbit is called _____ and the descending node _____.

Section B — Multiple choice ★ 10 marks

Circle the correct option.

1. Facing **north** at night from India, the stars appear to move:
   (a) clockwise around Polaris (b) anticlockwise around Polaris (c) straight upward (d) they do not move
2. Compared with last night, a given star rises tonight about:
   (a) 4 minutes later (b) 4 minutes earlier (c) 1 hour earlier (d) at the same time
3. The **main** cause of Earth's seasons is:
   (a) Earth's changing distance from the Sun (b) the tilt of Earth's axis (c) changes in the Sun's brightness (d) the Moon
4. On the day of an equinox, the Sun rises:
   (a) exactly due east (b) north of east (c) south of east (d) due north
5. Makar Sankranti is the day the Sun enters:
   (a) tropical Capricorn (b) sidereal Makara (c) Mesha (d) the summer solstice point
6. The ~24-day gap between the winter solstice and Makar Sankranti is caused by:
   (a) leap years (b) precession of the equinoxes (c) the Moon's orbit (d) time zones

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7. The **Full Moon** rises at about:
   (a) sunrise (b) noon (c) sunset (d) midnight
8. The time from one New Moon to the next (synodic month) is about:
   (a) 27.3 days (b) 29.5 days (c) 30.4 days (d) 28.0 days
9. Which of these is safe to watch with the **naked eye**?
   (a) a partial solar eclipse (b) an annular solar eclipse (c) a total lunar eclipse (d) all of these
10. Helium was discovered in the Sun's spectrum during the 1868 total solar eclipse observed from:
    (a) Guntur (b) Kolkata (c) Jaipur (d) Shimla

Section C — Match the following ★ 10 marks

C1. Match each term with its meaning (½ mark each, 4 marks)

Term	Meaning
1. Equinox	A. Dark central part of a shadow
2. Solstice	B. Day and night nearly equal
3. Perihelion	C. 18 years 11 days 8 hours eclipse cycle
4. Umbra	D. "Ring of fire" solar eclipse
5. Saros	E. Sun reaches its farthest north or south
6. Uttarayana	F. Earth's closest point to the Sun
7. Annular	G. Sun's six-month "northward journey"
8. Ecliptic	H. Sun's yearly path among the stars

Answers: 1–____ 2–____ 3–____ 4–____ 5–____ 6–____ 7–____ 8–____

C2. Festivals of the Sun. For each festival write the **state** (from the box) and whether it is celebrated at **Makar (M)** or **Mesha (Me)** Sankranti. (½ mark per cell, 6 marks)
States: Tamil Nadu · Gujarat · Punjab · Assam · Kerala · West Bengal

Festival	State	M or Me
Pongal		
Uttarayan (kite festival)		
Lohri (eve)		
Bohag (Rongali) Bihu		
Vishu		
Pohela Boishakh		

Section D — Short answers and patterns ★★ 12 marks

Answer in 2–4 sentences (2 marks each).

1. Why do we see **different star patterns in different seasons** in the evening sky?
2. What is a **Zero Shadow Day**? Why does **Delhi** never have one?
3. If the Moon goes round Earth every month, why don't we get a solar and a lunar eclipse **every** month?

4. **Complete the pattern.** The New Moon rises at about 6 AM. The First Quarter rises at about _____, the Full Moon at about _____ and the Last Quarter at about _____. In words, what is the rule?
5. Why does the Moon look **coppery red** during a total lunar eclipse instead of disappearing?
6. Rahul says, "It's hot in summer because the Earth is closer to the Sun." Give **two** pieces of evidence that prove him wrong.

Section E — Calculations ★★ 18 marks

Formula box

- Speed of rotation at latitude φ: $v \approx 1670 \cos\varphi$ km/h
- Stars rise ≈ 3.93 min (≈ 4 min) earlier each day; the Moon rises ≈ 50 min later each day
- Noon altitude of the Sun: $h = 90^\circ - \varphi + \delta$ (if $h > 90^\circ$, the Sun is north of overhead at $180^\circ - h$)
- Foucault pendulum: $T = 23.93\ \text{h} / \sin\varphi$

Worked example — study this first

Chennai is at 13.1° N. Noon altitude of the Sun on 21 June ($\delta = +23.4^\circ$):

$$h = 90^\circ - 13.1^\circ + 23.4^\circ = 100.3^\circ > 90^\circ \ \Rightarrow\ \text{north of overhead at } 180^\circ - 100.3^\circ = 79.7^\circ.$$

At noon, shadows point **south**.

E1. [2] How fast is a person in **Kolkata** ($\varphi = 22.6^\circ$) moving because of Earth's rotation? ($\cos 22.6^\circ = 0.923$)

E2. [3] Sirius rises at **9:30 PM on 1 October**. At about what time will it rise on **15 November** (45 days later)? Show working.

E3. [4] Calculate the noon altitude of the Sun: (a) Bengaluru (13.0° N) on 22 December ($\delta = -23.4^\circ$); (b) Delhi (28.6° N) on the March equinox; (c) Srinagar (34.1° N) on 21 June ($\delta = +23.4^\circ$); (d) which place/date has the **longest** noon shadow?

E4. [3] On **1 June** the Sun's declination is $+22.0^\circ$. Find the Sun's noon altitude in **Mumbai** (19.1° N). Is it north or south of overhead? Which way do noon shadows point?

E5. [3] Using the declination table in Lecture 1.2, estimate the two **Zero Shadow Days** of **Hyderabad** (17.4° N). (Hint: find the two months in which the Sun's declination passes 17.4°, then interpolate.)

E6. [2] Tonight the Moon rises at **7:10 PM**. Estimate when it will rise (a) tomorrow, (b) three days from now.

E7. [1] How long would a Foucault pendulum in **Delhi** (28.6° N) take to turn through a full circle? ($\sin 28.6^\circ = 0.479$)

Section F — Night-sky task ★★ 10 marks

Choose ONE and attach your record sheet.

F1. The 4-minute rule. Pick a bright star low in the east just after dark. Note the exact time it clears a fixed landmark (rooftop, tree, pole) from a fixed spot. Repeat **7 nights later** from the same spot. Record both times; compare the observed difference with the predicted $7 \times 4 = 28$ minutes, and give one reason why they might not match exactly.

F2. Polaris and latitude. Find Polaris using the Saptarishi pointers or the "W" (Lecture 1.1). Measure its altitude with your fist (1 fist $\approx 10^\circ$, 1 finger $\approx 1^\circ$) or a phone clinometer app. Compare with your city's latitude and suggest two sources of error.

F3. Sunset point. Photograph or sketch the sunset from the same spot on 3 days at least 1 week apart, marking its position against the skyline. Which way is the sunset point moving, and why?

Bonus challenge ★★★

up to 5 marks

Imagine Earth's axis had NO tilt (0°)**.** (a) Would India have seasons? (b) Would there be Zero Shadow Days in India? (c) How long would day and night be in Mumbai through the year? (d) Where would the Sun rise every day?